Explain the Linux file system hierarchy (/etc, /var/log, /home, /tmp) and file permissions: chmod, chown and SUID with examples.

Linux Basics & Permissions

File System Hierarchy

- `/etc`: Holds system configuration files (e.g., network settings, user lists).
- `/var/log`: Stores system, application, and security log files.
- `/home`: The personal directory for regular users to store their own files.
- `/tmp`: Stores temporary files that are usually wiped out when the system reboots.

Permissions & Commands

- **chmod**: Changes file permissions (Read=4, Write=2, Execute=1).
  - *Example*: `chmod 755 script.sh` (Owner can do everything; group and others can only read and execute).
- **chown**: Changes who owns a file or folder.
  - *Example*: `chown root:admin file.txt` (Sets user owner to root and group owner to admin).
- **SUID (Set Owner User ID)**: A special permission bit that lets a user run a file with the privileges of the file's owner (often root) instead of their own.
  - *The Risk*: If a binary like `find` or `bash` has the SUID bit set and is owned by root, a low-privileged user can abuse it to run root commands and instantly escalate their privileges.